

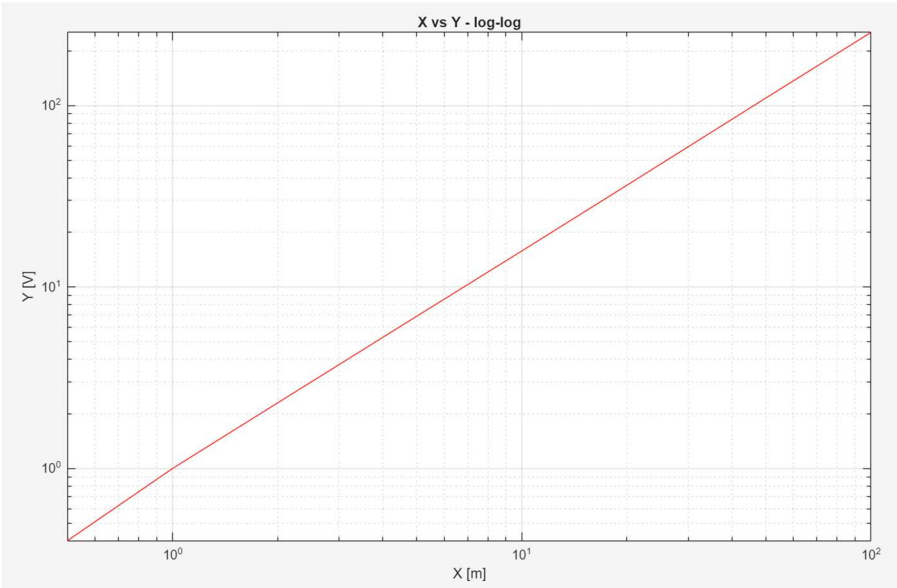
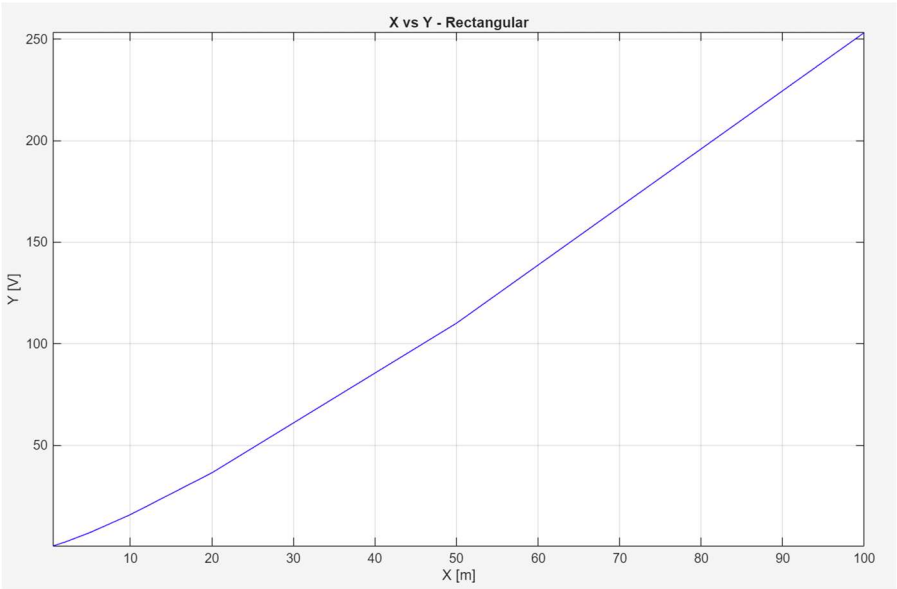
Day 5 – Calibration

NOTE: Claude was used for print-out generation (summaries) and troubleshooting.

Q1:

--- Table 1.5: Calibration Data Summary ---
X-Range . . . 0.50 to 100.00 [m]
X-Span . . . 99.50 [m]
Y-Range . . . 0.40 to 253.20 [V]
Y-Span . . . 252.80 [V]

Q2:



Logarithmic scale is best for larger scale, as it allows the interpreter to observe trends at lower and higher scales at the same time

Q3:

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--- Sensitivity Check ---
@ X = 5.0 . . . S = 1.688 [V/m]
@ X = 10.0 . . . S = 1.967 [V/m]
@ X = 20.0 . . . S = 2.357 [V/m]
The system is most sensitive at X = 20 (S = 2.357 [V/m])
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$$S_i = \frac{Y_{i+1} - Y_{i-1}}{X_{i+1} - X_{i-1}}$$

Q4:

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--- Table 1.7: Clock Calibration Results ---
Clock      Systematic[s]      Random[s]      Total[s]
A           144.000           1.000          144.003
B             5.000           0.000           5.000
C             0.000           1.732           1.732
```

Ranked by accuracy (best to worst): C B A

$$E_{sys} = avg(t_{diff}), E_{ran} = std(t_{diff}), E_{tot} = \sqrt{E_{sys}^2 + E_{ran}^2}$$

Q5:

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--- Table 1.6: Voltmeter Hysteresis Results ---
X[mV]      Y_inc[mV]      Y_dec[mV]      Hyst[mV]
0.0         0.10         0.20         0.10
1.0         1.10         1.20         0.10
2.0         2.10         2.20         0.10
3.0         3.00         3.20         0.20
4.0         4.10         4.20         0.10
5.0         5.00         5.00         0.00
```

Maximum Hysteresis. . . 0.20 [mV] at X = 3.0 [mV]

Maximum Hysteresis (%FS). . . 4.00 %

$$E_{hy} = Y_{dec} - Y_{inc}, \%FS = \frac{E_{hyst,max}}{range} * 100$$

Q6:

--- Table 1.1: Pressure Transducer Elemental Errors ---

Nominal Reading. . . 500.0 [cm H2O]

Temperature Range. . . 18.0 - 25.0 [degC] (dT = 7.0)

Error Source	Mag [cmH2O]
Linearity	5.000
Hysteresis	1.500
Sensitivity	1.250
Thermal Sensitivity	0.700
Thermal Zero Drift	1.400

Overall Instrument Error (RSS). . . 5.591 [cm H2O] (0.56 %FSO)

$$E_{lin} = \% * FSO, E_{hys} = \% * FSO, E_{sens} = \% * P, E_{tsen} = \% * P * dT, E_{tzero} = \% * FSO * dT$$